

LESSON 7.2

SLOPE IN REAL-WORLD CONTEXTS



LESSON INTRODUCTION

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

LEARNING TARGETS

- I can determine the slope from a graph, table, or written description in a real-world context.
- I can interpret the slope as the rate of change in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.2 Write a linear two-variable equation to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function and determine and interpret its key features.

ESSENTIAL QUESTIONS

- How do you find the rate of change from a graph, table, or written description?
- How do you interpret the slope as the rate of change in a real-world context?

LESSON NARRATIVE

Students will build on their knowledge of finding the rate of change using a table of values or a graph and apply it to interpreting the rate of change in a real-world context. Students will extend this knowledge to include how to recognize or determine the rate of change from a written description. Students will find and interpret the rate of change given any of the three representations of a linear relationship, graph, table of values, or written description.

COMPONENT SUMMARY

7.2.1 WARM-UP (~5 MIN.)

Graph the story from the video

7.2.2 GUIDED INSTRUCTION (10-15 MIN.)

Teacher models how to determine the rate of change from graphs, tables, and written descriptions of real-world situations

7.2.3 GUIDED PRACTICE (10 MIN.)

Calculate the rate of change and interpret it in real-world situations represented in different ways

7.2.4 YOUR TURN! (10 MIN.)

Practice determining and interpreting the rate of change from a graph and written description

7.2.5 WRAP-UP (~5 MIN.)

Find and interpret the rate of change from a table representing a real-world context

RESOURCES

GLOSSARY

Key Terms:

- Rate of change
- Slope

Additional Terms: N/A

MATERIALS

- Styrofoam cup stacking videos
- (Optional) Notecards
- (Optional) Graph paper

ADDITIONAL PREPARATION

- Warm-Up 7.2.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may still be confused on slope and think that $\text{slope} = \frac{\text{change in } x}{\text{change in } y}$.
- Students may struggle with staying consistent with their ordered pairs when finding the $\frac{\text{change in } y}{\text{change in } x}$.
- Students may struggle to recognize equivalent forms of numbers.
- Students may struggle to interpret the sign of the slope in real-world situations.
- Students may neglect to include the two units of measurement when expressing the rate of change in context.

THINGS TO CONSIDER

- Consider setting a timer for the 7.2.1 Warm-Up so it does not extend too long.
- Model for students to always include the units when they are available while expressing rate of change in context.
- Consider providing calculators for students who struggle with computations with the expectation that they document their calculations in the space provided. This way, students who may struggle with the computation can still focus on interpreting the meaning of slope in context.

LITERACY AND LANGUAGE ROUTINES

Algebra Talk

Consider using an Algebra Talk routine in the 7.2.3 Guided Practice. Have students share the different approaches they used to find the rate of change.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students determine and interpret the rate of change in real-world contexts. For example, in the 7.2.5 Wrap-Up, students must find the rate of change from a table of values and relate it back to the price of a gallon of gasoline.

DIFFERENTIATE INSTRUCTION

Consider having students create a reference card during 7.2.2 Guided Instruction with instructions and examples of finding the slope using a table, graph, and written description.

The graphs in 7.2.3 Guided Practice provide a visual entry point. The discussions in 7.2.2 Guided Instruction provide an auditory entry point.

7.2.1 WARM-UP: GRAPH THE STORY

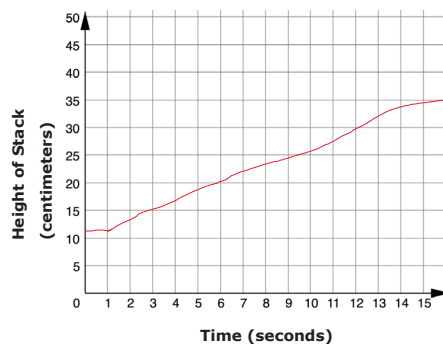
TEACHER MOVES

Ask students how they determined the points for their graph and what the points represent in context. Ask students to discuss with their partners whether or not the relationship between time, in seconds, and height of the stack, in centimeters, forms a straight line.



7.2.1 Warm-Up: Graph the Story

1. Watch the video of the person stacking styrofoam cups. Create a sketch of the graph to represent the height of the stack, in centimeters, over time.



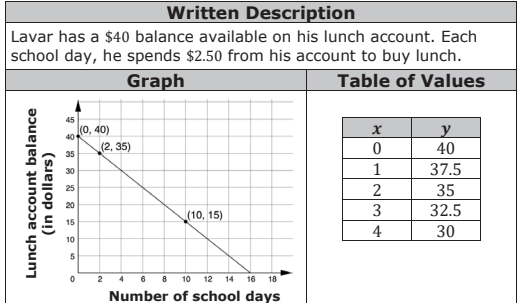
Answers will vary. Sample response shown on the graph.



7.2.2 Guided Instruction: Finding and Interpreting Slope in Context

A linear relationship is any relationship between two quantities, where one quantity has a constant rate of change with respect to the other. This means that one quantity is increasing or decreasing the same amount for every equal-sized interval of the other quantity.

Linear relationships can be used to model real-world contexts in which the quantities are changing at a constant rate. Three different representations of the same linear relationship are shown.



1. Use the graph to find the slope of the line.

$$\frac{40 - 35}{0 - 2} = \frac{5}{-2} = \frac{-5}{2} = -\$2.50 \text{ per day}$$
2. Use the table to find the slope of the line.

$$\frac{40 - 37.5}{0 - 1} = \frac{2.5}{-1} = -\$2.50 \text{ per day}$$

7.2.2 GUIDED INSTRUCTION: FINDING AND INTERPRETING SLOPE IN CONTEXT

DIFFERENTIATE INSTRUCTION

During instruction, provide additional scaffolding by underlining and circling words in the written description to help students recognize how to determine the slope from a written description. Consider modeling questions 1 and 2 using a specific color relating the written work to the graph and table of values. Then use the same color to indicate how to find the slope from the written description.

TEACHER MOVES

In certain contexts (especially ones dealing with money), rate of change can, and sometimes should, be represented as a decimal with two decimal places to better interpret the rate of change with cost.

TEACHER MOVES

Consider having students create a reference card with instructions and examples of finding the slope using a table, graph, and written description.

7.2.2 GUIDED INSTRUCTION: FINDING AND INTERPRETING SLOPE IN CONTEXT (CONTINUED)

ENRICHMENT

If Lavar is spending money, will the balance be increasing or decreasing? How does this relate to the sign of the slope?

ENRICHMENT

If Tabitha is charging customers \$15 per hour, will the amount of money she earns increase or decrease? How does this relate to the sign of the slope?

DIFFERENTIATE INSTRUCTION

Consider allowing students to create a graphic organizer or index card detailing the definition and examples of slope. Students can use the definition of slope and its relationship to rate of change as a guide. Encourage students to use their graphics as an anchor chart throughout the lesson.

The written description of the relationship can also be used to find the slope.

3. Use the written description to complete the statements describing how to find the slope.

The balance of Lavar's lunch account is ☐ increasing ☒ decreasing

each school day, so the slope is ☐ positive. ☒ negative. It

changes by ☐ \$40 ☒ \$2.50 each school day, so the slope is

☐ 2.5.
☒ -2.5.
☐ 40.
☐ -40.

Tabitha earns money mowing lawns each weekend. She charges customers \$15 per hour to mow their lawn, in addition to the \$7 flat fee she charges them for gasoline and other supplies she'll need.

4. Two values are given in the context. Explain which value represents the rate of change.
\$15 per hour represents the rate of change because it is a rate comparing two quantities (money and time).
5. Determine whether the rate of change represents an increase or decrease in the amount of money Tabitha earns. Show your work or explain your answer.
The rate of change represents an increase since she earns \$15 more every hour she works.
6. Based on the answers to the previous questions, what is the slope in this linear relationship?
$$\frac{\$15}{1 \text{ hour}} = 15$$

7. Complete the statement to interpret the slope as the rate of change for this context.

In this context, the slope means that the amount of

money Tabitha earns ☐ increases ☒ decreases by ☐ 7 ☒ 15

☐ dollars ☒ hours for every ☐ dollar ☒ hour she mows a customer's lawn.

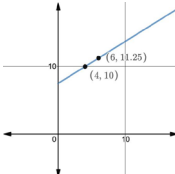


The **slope** is the **rate of change** that one quantity is changing by in terms of another quantity. When interpreting the rate of change in a real-world context, **units for both quantities** are included.



7.2.3 Guided Practice: Relating Slope to Rate of Change in Real-World Context

For each linear relationship, determine the rate of change and interpret its meaning in the given context.

Linear Relationship	Rate of Change	Interpretation								
The table of values represents the relationship between the cost, C, in dollars of s, bags of sour patch candies. <table border="1"><thead><tr><th>s</th><th>C</th></tr></thead><tbody><tr><td>2</td><td>2.5</td></tr><tr><td>3</td><td>3.75</td></tr><tr><td>4</td><td>5</td></tr></tbody></table>	s	C	2	2.5	3	3.75	4	5	Work may vary. $\frac{2.5 - 3.75}{2 - 3} = \frac{-1.25}{-1} = 1.25$	The cost is increasing \$1.25 per bag of sour patch candies. OR The candy costs \$1.25 per bag.
s	C									
2	2.5									
3	3.75									
4	5									
The graph represents the relationship between the average baby's weight, y, in pounds and a baby's age, x, in months. 	Work may vary. $\frac{11.25 - 10}{6 - 4} = \frac{1.25}{2} = 0.625$	The average baby's weight increases 0.625 pounds each month.								
A mountain road gains elevation at a constant rate. After 2 miles, the elevation is 5,500 feet above sea level. After 4 miles, the elevation is 6,200 feet above sea level.	Work may vary. $\frac{5500 - 6200}{2 - 4} = \frac{-700}{-2} = 350$	The road's elevation increases 350 feet every mile.								

7.2.3 GUIDED PRACTICE: RELATING SLOPE TO RATE OF CHANGE IN REAL-WORLD CONTEXT

TEACHER MOVES

Consider using an Algebra Talk routine to have students share the different approaches they used to find the rate of change.

DIFFERENTIATE INSTRUCTION

The graph provides a visual entry point for students and makes it clear whether the rate of change is increasing or decreasing. Provide additional entry points for students by providing opportunities for them to discuss their thinking when finding and interpreting the rate of change from a table or written description.

7.2.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider placing students in pairs to discuss their answers after they first try the questions individually. The discussion will build student confidence and understanding. For students struggling with the computation, consider allowing them to use calculators to find the slope so they can practice interpreting as well.



7.2.4 Your Turn!

For each linear relationship, determine the rate of change and interpret its meaning in the given context.

Linear Relationship	Rate of Change	Interpretation
The graph shows the height in inches, h, of a bamboo plant t months after it has been planted.	<i>Work may vary.</i> $\frac{30 - 15}{6 - 1} = \frac{15}{5} = 3$	<i>The bamboo plant's height increases 3 inches each month.</i>
Lin is reading a book for an assignment in English. After reading the first 30 pages, she makes a plan to read 40 pages each day until she finishes the book.	<i>Work may vary.</i> $\frac{40 \text{ pages}}{1 \text{ day}}$	<i>Lin reads 40 pages every day.</i>



7.2.5 Wrap-Up: Ticket Out the Door

Yingjie tracked the cost of purchasing gasoline and the number of gallons he bought. His table is shown where x represents the number of gallons purchased, and y represents the cost in dollars.

1. What is the rate of change?

$$\frac{19.71 - 21.9}{5.4 - 6} = \frac{-2.19}{-0.6} = 3.65$$

2. Interpret the rate of change.

Gasoline costs \$3.65 a gallon.

x	y
5.4	19.71
6	21.90
15	54.75
17	62.05

7.2.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

If students are struggling with computation, consider allowing students to use calculators. This will help you determine if students are struggling with the computation only or if they do not understand the concept of rate of change.